

# TrES-3b

R-0402

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_180615 · created 2026-07-19T04:15:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2458284.726 +/- 0.0014 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.194 +/- 0.057                |
| Transit depth (Rp/Rs)^2                           | 3.76 +/- 2.21 [%]              |
| Orbital Inclination (inc)                         | 86.85 +/- 2.91                 |
| Airmass coefficient 1 (a1)                        | 1.99 +/- 0.9                   |
| Airmass coefficient 2 (a2)                        | -0.52 +/- 0.66                 |
| Scatter in the residuals of the lightcurve fit is | 47.91 %                        |
| Best Comparison Star                              | #1 - [278, 129]                |
| Optimal Aperture                                  | 2.82                           |
| Optimal Annulus                                   | 7.9                            |
| Transit Duration (day)                            | 0.08 +/- 0.012                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                         |
|--------------------------------------|---------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.194 $\pm$ 0.057 vs 0.16309 (deviation 0.54 $\sigma$ ) |
| Duration fit vs published            | 0.08 d vs 0.0567 d (deviation 1.94 $\sigma$ )           |
| Mid-time O–C                         | 2.6 min                                                 |
| Depth SNR                            | 3.4                                                     |
| Residual scatter                     | 47.91 %                                                 |

## Light curve

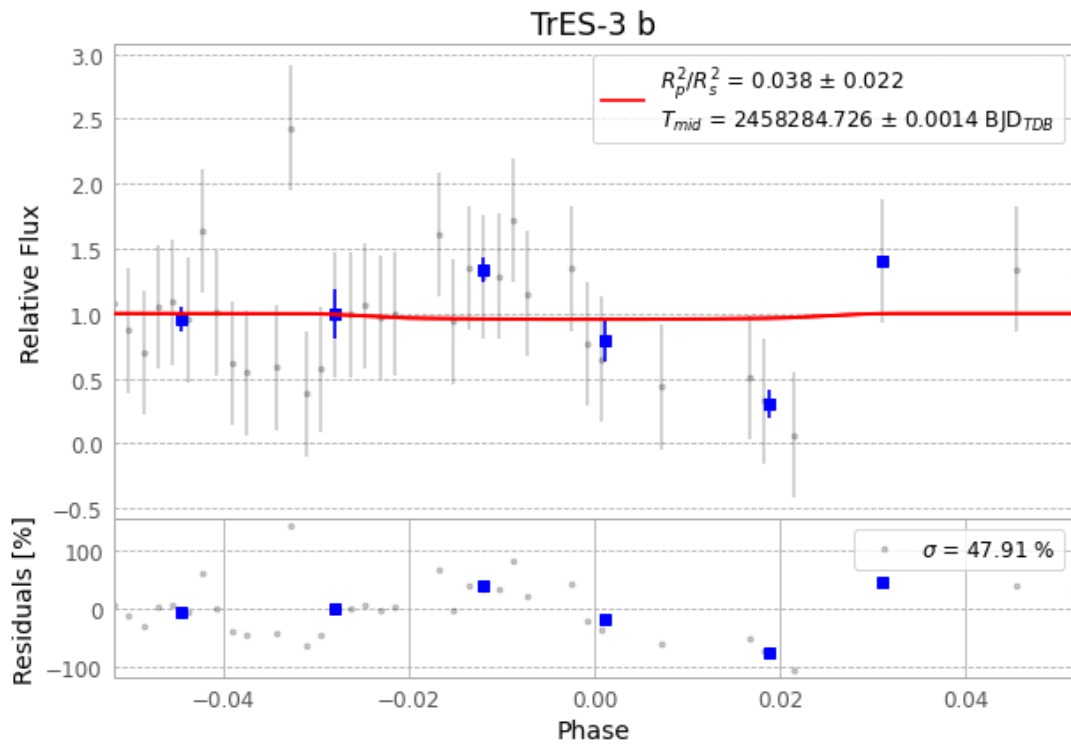


Figure 1 — FinalLightCurve\_TrES-3 b\_2018-06-14.png (SHA-256 6fd14ecea6051615...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 175], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 6a27bea07e7086c3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 39d9b80

## Data provenance

67 FITS frames (44 MB), TRES-3180615034212.FITS → TRES-3180615070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180615.log (461656745525...), FinalLightCurve\_TrES-3 b\_2018-06-14.png (6fd14ecea605...), FinalLightCurve\_TrES-3 b\_2018-06-14.csv (58cd1b5e321f...)

## Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 04:15Z.